30 40 50 60

−6 −4 −2 0

ax − 2002

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040 0.045

bx1 − 2002

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−0.05 0.00 0.05 0.10

bx2 − 2002

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1950 1960 1970 1980 1990 2000

−150 −100 −50 0 50

kt − 2002

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−100 0 100 200 300 400 500

gc − 2002

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2003

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−50 −40 −30 −20 −10 0

ax − 2003

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2003

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−0.04 −0.02 0.00 0.02 0.04 0.06 0.08

bx2 − 2003

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1950 1960 1970 1980 1990 2000

−2000 −1000 0 1000 2000

kt − 2003

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−4000 −3000 −2000 −1000 0 1000 2000 3000

gc − 2003

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2004

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−80 −60 −40 −20 0

ax − 2004

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2004

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−0.04 −0.02 0.00 0.02 0.04 0.06 0.08

bx2 − 2004

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1950 1960 1970 1980 1990 2000

−2000 −1000 0 1000 2000 3000 4000

kt − 2004

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−10000 −5000 0

gc − 2004

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2005

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−60 −40 −20 0

ax − 2005

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2005

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−0.01 0.00 0.01 0.02 0.03 0.04 0.05

bx2 − 2005

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1950 1960 1970 1980 1990 2000

−3000 −2000 −1000 0 1000 2000 3000

kt − 2005

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−4000 −2000 0 2000 4000

gc − 2005

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2006

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−60 −40 −20 0

ax − 2006

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2006

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−0.02 0.00 0.02 0.04 0.06

bx2 − 2006

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−3000 −2000 −1000 0 1000 2000 3000

kt − 2006

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−4000 −2000 0 2000 4000

gc − 2006

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2007

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−50 −40 −30 −20 −10 0 10

ax − 2007

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2007

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−0.02 0.00 0.02 0.04 0.06

bx2 − 2007

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−2000 −1000 0 1000 2000

kt − 2007

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−2000 0 2000 4000

gc − 2007

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2008

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−50 −40 −30 −20 −10 0 10

ax − 2008

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2008

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx2 − 2008

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−2000 −1000 0 1000 2000

kt − 2008

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−4000 −2000 0 2000 4000

gc − 2008

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2009

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−50 −40 −30 −20 −10 0 10

ax − 2009

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2009

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx2 − 2009

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−2000 −1000 0 1000 2000

kt − 2009

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−2000 0 2000 4000

gc − 2009

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2010

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−30 −20 −10 0

ax − 2010

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2010

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2010

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−2000 −1000 0 1000

kt − 2010

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−2000 −1000 0 1000 2000 3000

gc − 2010

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2011

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−8 −7 −6 −5 −4 −3

ax − 2011

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2011

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.005 0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2011

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−150 −100 −50 0 50 100 150

kt − 2011

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−300 −200 −100 0 100 200

gc − 2011

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2012

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−10 −5 0 5

ax − 2012

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2012

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.005 0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx2 − 2012

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−500 0 500 1000

kt − 2012

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−1500 −1000 −500 0 500 1000

gc − 2012

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2013

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−10 −5 0 5

ax − 2013

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2013

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.005 0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2013

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−500 0 500 1000

kt − 2013

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−1500 −1000 −500 0 500 1000

gc − 2013

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2014

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−8 −7 −6 −5 −4 −3

ax − 2014

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx1 − 2014

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.005 0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2014

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−150 −100 −50 0 50 100 150

kt − 2014

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−100 0 100 200

gc − 2014

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2015

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−10 −5 0 5 10

ax − 2015

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04

bx1 − 2015

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.005 0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2015

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−1000 −500 0 500 1000 1500

kt − 2015

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−2000 −1000 0 1000

gc − 2015

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2016

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−9 −8 −7 −6 −5 −4 −3

ax − 2016

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx1 − 2016

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2016

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000

−200 −100 0 100 200

kt − 2016

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−200 −100 0 100 200 300

gc − 2016

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2017

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−10 −5 0 5 10

ax − 2017

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx1 − 2017

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2017

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000 2020

−1000 −500 0 500 1000 1500

kt − 2017

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−2000 −1000 0 1000

gc − 2017

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2018

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−12 −10 −8 −6 −4 −2

ax − 2018

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx1 − 2018

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035

bx2 − 2018

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000 2020

−400 −200 0 200 400

kt − 2018

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−400 −200 0 200 400 600 800

gc − 2018

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012

mx_fc − 2019

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−10000 −8000 −6000 −4000 −2000 0

ax − 2019

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.01 0.02 0.03 0.04 0.05

bx1 − 2019

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.00 0.01 0.02 0.03

bx2 − 2019

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000 2020

−1500 −1000 −500 0 500 1000 1500

kt − 2019

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980

−3e+07 −2e+07 −1e+07 0e+00

gc − 2019

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0 500 1000 1500 2000

ax − 2020

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040 0.045

bx1 − 2020

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.00 0.01 0.02 0.03 0.04

bx2 − 2020

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000 2020

−1000 −500 0 500 1000

kt − 2020

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980 2000

0e+00 1e+06 2e+06 3e+06 4e+06

gc − 2020

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2021

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−10 0 10 20

ax − 2021

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.015 0.020 0.025 0.030 0.035

bx1 − 2021

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.010 0.015 0.020 0.025 0.030 0.035 0.040

bx2 − 2021

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000 2020

−1500 −1000 −500 0 500 1000 1500

kt − 2021

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980 2000

−2000 −1000 0 1000 2000

gc − 2021

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014 0.016

mx_fc − 2022

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

−7 −6 −5 −4

ax − 2022

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.015 0.020 0.025 0.030 0.035 0.040

bx1 − 2022

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.00 0.01 0.02 0.03 0.04

bx2 − 2022

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1940 1960 1980 2000 2020

−60 −40 −20 0 20 40 60

kt − 2022

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

1880 1900 1920 1940 1960 1980 2000

0 100 200 300

gc − 2022

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast

30 40 50 60

0.002 0.004 0.006 0.008 0.010 0.012 0.014

mx_fc − 2023

Converged + Forecast
Converged + No Forecast
Not Converged + No Forecast